

Data Challenge NLP

NLP COIN APP

Ist eine WebApp zur Analyse und Visualisierung von Ground Truth Daten.

Aufgebaut mit Python  und Streamlit 

Features

- Vergleich zwischen Modellvorhersagen und der Ground Truth
- Generierung einer Modellvorhersage für eine Münzbeschreibung oder Testdatensatz
- Vergleich zwischen zwei Ground Truth Datensätzen und graphische Darstellung der Beziehungen
- Die Bestimmung der Genauigkeit von Wörtern sowie von den Beziehungen

Erkenntnisse

Fehlertypen

- Fehlende Ground Truth für bestimmte Designs
- Uneinheitliche Zuordnung semantisch ähnlicher Wörter
- Leere Vorhersagen
- Doppelte Relationen in den Vorhersagen
- Fehlerhafte Entitäten und/oder Relationen

Beispiel

Design 570

Confronted busts of Gordian III, laureate, wearing paludamentum, and of Tranquillina, diademed and draped.

Prediction

[(` Gordian`, ` PERSON`, ` wearing`, ` paludamentum`, ` OBJECT`)]

Ground Truth

[(` Gordian`, ` PERSON`, ` wearing`, ` cuirass`, ` OBJECT`)]

Metriken

category	all	from CN	from OCRE	from RE_yaml	↓ auto_generated
Precision	96.47	95.87	0	98.57	98.36
F1	65.44	63.78	0	63.3	88.24
Recall	49.52	47.79	0	46.62	80
Precision (scikit-learn)	42.9	43.13	0	49.04	40.33
Recall (scikit-learn)	42.94	43.08	0	49.04	40.33
F1 Score (scikit-learn)	42.89	43.08	0	49.04	40.33
Jaccard Index	28.41	28.84	0	36.21	26.78
Hamming Loss	2.52	2.78	0	7.08	13.88
Subset Accuracy	0.12	0.86	0	9.59	1.64

Demo

NLP-COIN-APP: HOME

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NLP COIN APP

Data Challenge SS24, Saif Samir & Luisa Cardenas

The NLP COIN APP was developed to provide support in the analysis and visualization of the Ground Truth Data. The analyses are useful for detecting errors in the Ground Truth and for comparing different approaches to create a new Ground Truth.

Features

- The comparison between model predictions and the ground truth, as well as the generation of a prediction for a coin description or the test dataset.
- The comparison between two ground truth datasets and the graphical display of the relationships.
- To determine the accuracy of words as well as the relationships between ground truth data.
- Additional insights we have gained.

NLP-COIN-APP: GT-PRED-COMPARISON

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Prediction - Ground Truth Comparison

Choose CSV files

 Drag and drop files here
Limit 200MB per file • CSV

Browse files

Single Prediction

With this feature, you can generate a prediction for your coin description. Just insert the description in the field and click enter.

Type the description to get the prediction.

▶ []

Prediction for Test Data

This feature allows you to generate predictions for test data. Simply click the button below.

Prediction for test data

NLP-COIN-APP: GT-PRED-COMPARISON

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Performance

Here is the data you uploaded for ground truth:

Unnamed: 0	design_id	y
32	1,108	445
33	2,324	28,053
34	2,194	27,748
35	769	2,267
36	501	1,347
37	1,299	1,966
38	431	24,798
39	1,583	4,276
40	104	1,711
41	2,056	1,212

Compare Ground Truth and Prediction

Calculation of the difference between prediction and ground truth

Now you can compare your uploaded prediction data against the uploaded ground truth data.

	design_id	prediction	ground_truth	not_in_ground_truth	missing_predictions	design_en
19	118		Artemis,PERSON,wearing,garment,OBJECT	Artemis,PERSON,holding,arrow,OBJE	Artemis,PERSON,holding,torch,OBJECT	Artemis standing facing, head left, wearing long garment, holding an arrow in right hand
20	121		Asclepius,PERSON,resting_on,serpent,ANIMAL		Asclepius,PERSON,resting_on,serpent,ANIMAL	Asclepius reclining left, on winged serpent to right.
21	135		Athena,PERSON,wearing,helmet,OBJECT		Athena,PERSON,wearing,helmet,OBJECT	Head of Athena, right, wearing crested Corinthian helmet, decorated with serpent, a
22	136	Athena,PERSON,wearing,helmet,OBJECT	Athena,PERSON,wearing,helmet,OBJECT			Head of Athena, right, wearing crested Corinthian helmet.
23	144	Athena,PERSON,holding,Nike,PERSON	Athena,PERSON,resting_on,shield,OBJEC	Athena,PERSON,holding,Nike,PERSON	Athena,PERSON,resting_on,shield,OBJEC	Athena (Nikephoros) enthroned left, holding Nike in outstretched right hand; left arm
24	269	Athena,PERSON,holding,Nike,PERSON	Athena,PERSON,resting_on,shield,OBJEC	Athena,PERSON,holding,Nike,PERSON	Athena,PERSON,resting_on,shield,OBJEC	Athena (Nikephoros) enthroned left, holding Nike in outstretched right hand; left arm
25	170	Athena,PERSON,holding,Nike,PERSON	Athena,PERSON,resting_on,shield,OBJEC	Athena,PERSON,holding,Nike,PERSON	Athena,PERSON,resting_on,shield,OBJEC	Athena (Nikephoros) enthroned left, holding Nike in outstretched right hand; left arm
26	182		Athena,PERSON,wearing,helmet,OBJECT	Athena,PERSON,holding,Nike,PERSON	Athena,PERSON,holding,spear,OBJECT	Athena standing facing, head left, wearing Corinthian helmet, holding Nike in outstre
27	187		Athena,PERSON,holding,spear,OBJECT	Athena,PERSON,resting_on,shield,OBJEC	Athena,PERSON,resting_on,shield,OBJEC	Athena standing right, holding spear in right hand, resting left arm on shield set on th
28	191		Athena,PERSON,wearing,garment,OBJECT	Athena,PERSON,wearing,helmet,OBJ	Athena,PERSON,holding,patera,OBJECT	Athena standing facing, head left, wearing long garment and helmet; holding patera i

Single Prediction

With this feature, you can generate a prediction for your coin description. Just insert the description in the field and click enter.

Type the description to get the prediction.

* []

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NLP-COIN-APP: GT-COMPARISON

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Ground Truth Prediction Comparison

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Ground Truth Comparison

Choose CSV files

Drag and drop files here

Limit 200MB per file + CSV

Browse files

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Ground Truth Comparison

Choose CSV files

Drag and drop files here

Limit 200MB per file + CSV

Browse files

second_ground_truth.csv

337.9KB

×

first_ground_truth.csv

362.9KB

×

Files uploaded successfully!

Here are your uploaded files. You can now perform various actions on your data.

Number of rows first ground truth: 2390

Here is the first Ground Truth:

	Unnamed: 0	design_id	y
0	0	6,652	[[('Aequitas', 'PERSON', 'wearing', 'chiton', 'OBJECT'), ('Aequitas', 'PERSON', 'wearing', 'chiton', 'OBJECT')]]
1	1	6,652	[[('Aequitas', 'PERSON', 'wearing', 'chiton', 'OBJECT'), ('Aequitas', 'PERSON', 'wearing', 'chiton', 'OBJECT')]]
2	2	6,652	[[('Aequitas', 'PERSON', 'wearing', 'chiton', 'OBJECT'), ('Aequitas', 'PERSON', 'wearing', 'chiton', 'OBJECT')]]
3	3	6,652	[[('Aequitas', 'PERSON', 'wearing', 'chiton', 'OBJECT'), ('Aequitas', 'PERSON', 'wearing', 'chiton', 'OBJECT')]]
4	4	9	[[('amphora', 'OBJECT', 'holding', 'poppy', 'PLANT'), ('amphora', 'OBJECT', 'holding', 'cc')]]
5	5	9	[[('amphora', 'OBJECT', 'holding', 'poppy', 'PLANT'), ('amphora', 'OBJECT', 'holding', 'cc')]]
6	6	1,856	[[('Anubis', 'PERSON', 'holding', 'palm branch', 'OBJECT')]]
7	7	1,582	[[('Aphrodite', 'PERSON', 'wearing', 'chiton', 'OBJECT'), ('Aphrodite', 'PERSON', 'holding', 'apple', 'PLANT')]]
8	8	1,582	[[('Aphrodite', 'PERSON', 'wearing', 'chiton', 'OBJECT'), ('Aphrodite', 'PERSON', 'holding', 'apple', 'PLANT')]]
9	9	24,725	[[('Apollo', 'PERSON', 'holding', 'patera', 'OBJECT'), ('Apollo', 'PERSON', 'holding', 'laurel', 'PLANT')]]

Number of rows second ground truth: 2420

Here is the second Ground Truth:

	design_id	y
0	27	[[('Antoninus Pius', 'PERSON', 'wearing', 'cultrass', 'OBJECT'), ('Antoninus Pius', 'PERSON', 'wearing', 'cultrass', 'OBJECT')]]
1	28	[[('Antoninus Pius', 'PERSON', 'wearing', 'paludamentum', 'OBJECT')]]
2	33	[[('Aphrodite', 'PERSON', 'holding', 'apple', 'PLANT')]]
3	36	[[('Aphrodite', 'PERSON', 'holding', 'breast', 'OBJECT'), ('Aphrodite', 'PERSON', 'holding', 'breast', 'OBJECT')]]

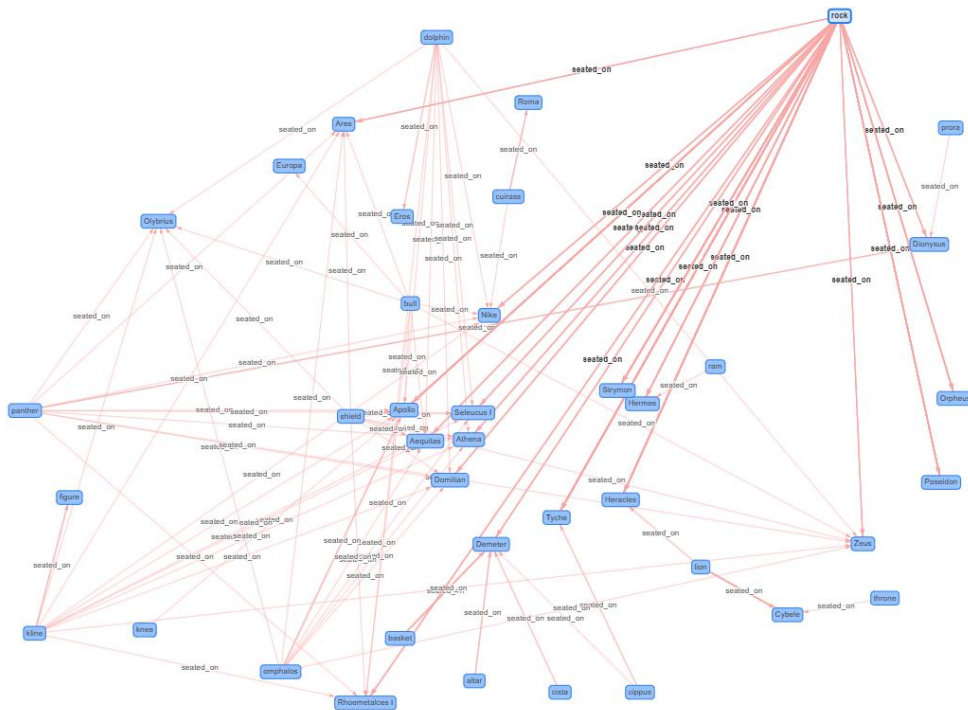
NLP-COIN-APP: GT-COMPARISON

[illegible]

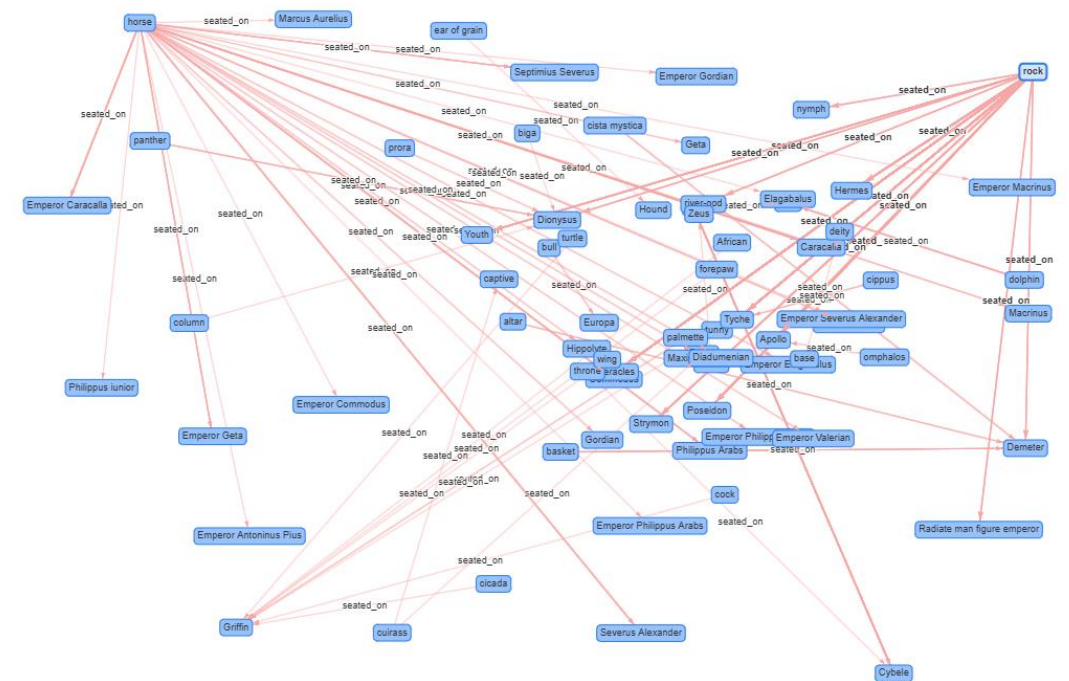
NLP-COIN-APP: Example: Relation seated_on

First GT

Knowledge Graph



Second GT



NLP-COIN-APP: METRICS

NLP COIN APP

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Analyse Tool

choose your analyse option:

You selected: Relation

Select left column

first_ground_truth

Relation

Relation

Word

Select right column

second_ground_truth

	design_id	relation	first_ground_truth	second_ground_truth	Diff	Acc
0	9	amphora OBJECT holding corn PLANT	1	0	1	
1	9	amphora OBJECT holding poppy PLANT	1	0	1	
2	10	Anchialos PERSON wearing taenia OBJECT	1	0	1	
3	24	Antoninus Pius PERSON wearing cuirass OBJECT	1	0	1	
4	24	Antoninus Pius PERSON wearing paludamentum OBJECT	1	0	1	
5	27	Antoninus Pius PERSON wearing cuirass OBJECT	1	1	0	1
6	27	Antoninus Pius PERSON wearing paludamentum OBJECT	1	1	0	1
7	28	Antoninus Pius PERSON wearing paludamentum OBJECT	1	1	0	1
8	33	Aphrodite PERSON holding apple PLANT	1	1	0	1
9	36	Aphrodite PERSON holding breast OBJECT	0	1	-1	0

Total results

	relation	design_id	first_ground_truth	second_ground_truth	Diff	Acc
0	Aequitas PERSON feeding dolphin ANIMAL	27703	1.0	0.0	1.0	inf
1	Aequitas PERSON feeding panther ANIMAL	27549	1.0	0.0	1.0	inf
2	Aequitas PERSON holding bow OBJECT	27811	1.0	0.0	1.0	inf
3	Aequitas PERSON holding club OBJECT	27621	1.0	0.0	1.0	inf
4	Aequitas PERSON holding cornucopia OBJECT	2547,6652	1.0	1.0	0.0	1.0
5	Aequitas PERSON holding cubit role OBJECT	2280,2542,3445	0.0	3.0	-3.0	0.0
6	Aequitas PERSON holding jar OBJECT	27858	1.0	0.0	1.0	inf
7	Aequitas PERSON holding paludamentum OBJECT	27637	1.0	0.0	1.0	inf

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NLP-COIN-APP: PERFORMANCE

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Metrics

Suche category

category	all	from CN	from OCRE	from RE_yaml	auto_generated
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Subset Accuracy	0.12	0.86	0	9.59	1.64
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Recall (scikit-learn)	42.94	43.08	0	49.04	40.33
F1 Score (scikit-learn)	42.89	43.08	0	49.04	40.33
Precision	96.47	95.87	0	98.57	98.36
Recall	49.52	47.79	0	46.62	80
F1	65.44	63.78	0	63.3	88.24

Missed Predictions

	Unnamed: 0	design_id	prediction	ground_truth	design_en	not_in_ground_truth_predictions	missing_predictions
8	8	665	[('Heracles', 'PERSON', 'wearing', 'lion skin', 'OBJECT')]	[('Heracles', 'PERSON', 'wearing', 'lion skin', 'OBJECT'), ('Heracles', 'PERSON', 'grasping', 'Hydra', 'OBJECT')]	Heracles standing left, wearing lion skin, grasping Hydra colling around his right leg w	set()	[('Heracles', 'PERSON', 'grasping', 'Hydra', 'OBJECT')]
9	9	672	[('Heracles', 'PERSON', 'resting_on', 'rock', 'OBJECT'), ('Heracles', 'PERSON', 'seated_on', 'rock', 'OBJECT')]	[('Heracles', 'PERSON', 'seated_on', 'rock', 'OBJECT'), ('rock', 'OBJECT', 'wearing', 'lion skin', 'OBJECT')]	Nude Heracles seated left on rock, head and torso facing, rock covered with a lion ski	[('Heracles', 'PERSON', 'resting_on', 'rock', 'OBJECT')]	set()
10	10	673	[('Heracles', 'PERSON', 'resting_on', 'rock', 'OBJECT'), ('Heracles', 'PERSON', 'resting_on', 'club', 'OBJECT')]	[('Heracles', 'PERSON', 'seated_on', 'rock', 'OBJECT'), ('rock', 'OBJECT', 'wearing', 'lion skin', 'OBJECT')]	Nude Heracles seated left on rock covered with lion skin, extending right hand, restin	set()	[('Heracles', 'PERSON', 'resting_on', 'club', 'OBJECT')]
11	11	674	[('Heracles', 'PERSON', 'resting_on', 'rock', 'OBJECT'), ('Heracles', 'PERSON', 'resting_on', 'club', 'OBJECT')]	[('Heracles', 'PERSON', 'seated_on', 'rock', 'OBJECT'), ('rock', 'OBJECT', 'wearing', 'lion skin', 'OBJECT')]	Nude Heracles seated left on rock covered with lion skin, resting right arm on club, re	set()	[('Heracles', 'PERSON', 'resting_on', 'club', 'OBJECT')]
12	12	679	[('Heracles', 'PERSON', 'holding', 'torch', 'OBJECT'), ('Heracles', 'PERSON', 'holding', 'club', 'OBJECT')]	[('Heracles', 'PERSON', 'holding', 'torch', 'OBJECT'), ('Heracles', 'PERSON', 'holding', 'club', 'OBJECT')]	Heracles advancing left, holding transverse lit torch in outstretched right hand and cl	set()	set()
13	13	711	[('Hermes', 'PERSON', 'resting_on', 'rock', 'OBJECT'), ('Hermes', 'PERSON', 'wearing', 'petasus', 'OBJECT')]	[('Hermes', 'PERSON', 'wearing', 'petasus', 'OBJECT'), ('Hermes', 'PERSON', 'holding', 'caduceus', 'OBJECT')]	Nude Hermes seated left on rock, wearing petasus, holding caduceus in right hand ar	[('Hermes', 'PERSON', 'resting_on', 'rock', 'OBJECT')]	set()
14	14	738	[('Homonoia', 'PERSON', 'wearing', 'kalathos', 'OBJECT'), ('Homonoia', 'PERSON', 'holding', 'corn', 'OBJECT')]	[('Demeter', 'PERSON', 'holding', 'corn', 'PLANT'), ('Homonoia', 'PERSON', 'wearing', 'kalathos', 'OBJECT')]	Demeter and Homonoia; inbetween, altar; to left, Demeter standing right, holding ea	set()	[('Demeter', 'PERSON', 'holding', 'corn', 'PLANT')]
15	15	852	[('figure', 'PERSON', 'seated_on', 'kline', 'OBJECT'), ('figure', 'PERSON', 'seated_on', 'kline', 'OBJECT')]	[('figure', 'PERSON', 'resting_on', 'kline', 'OBJECT'), ('figure', 'PERSON', 'seated_on', 'kline', 'OBJECT')]	Banquet scene; male figure reclining left on kline and placing his right hand on the le	set()	[('figure', 'PERSON', 'resting_on', 'kline', 'OBJECT')]
16	16	854	[('figure', 'PERSON', 'seated_on', 'kline', 'OBJECT'), ('figure', 'PERSON', 'seated_on', 'kline', 'OBJECT')]	[('figure', 'PERSON', 'resting_on', 'kline', 'OBJECT'), ('figure', 'PERSON', 'seated_on', 'kline', 'OBJECT')]	Banquet scene; male figure reclining left on kline, placing his right hand on the left sh	set()	[('figure', 'PERSON', 'resting_on', 'kline', 'OBJECT')]
17	17	908	[('Nike', 'PERSON', 'holding', 'trophy', 'OBJECT')]	[('Nike', 'PERSON', 'holding', 'trophy', 'OBJECT'), ('Nike', 'PERSON', 'seated_on', 'cuirass', 'OBJECT')]	Nike advancing right, holding transverse trophy in both hands over bound captive se	set()	[('Nike', 'PERSON', 'seated_on', 'cuirass', 'OBJECT')]

Empty Table